Human Activity Recognition

DATS 6303 – Deep Learning

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Introduction

What is HAR?

The task of identifying and classifying human actions from video sequences using computer vision and deep learning.

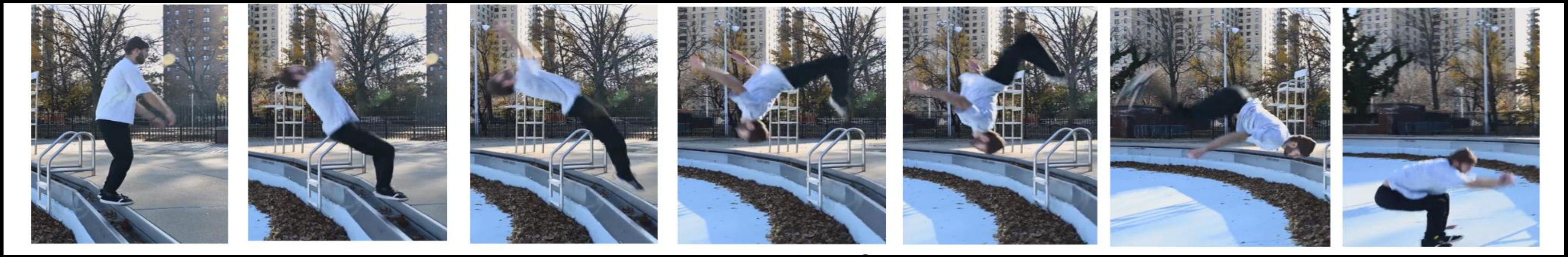
Why is HAR Important?

- **Security**: Detects suspicious activities in real-time (e.g., falls, fights).
- **Sports Analytics**: Tracks player movements and performance.
- **Human-Computer Interaction**: Enables gesture-based controls and adaptive systems.

Problem Statement

To Classify activities from videos.

Key Challenges:
- Varying lighting and camera angles
- Complex, overlapping movements
- Environmental clutter and occlusion

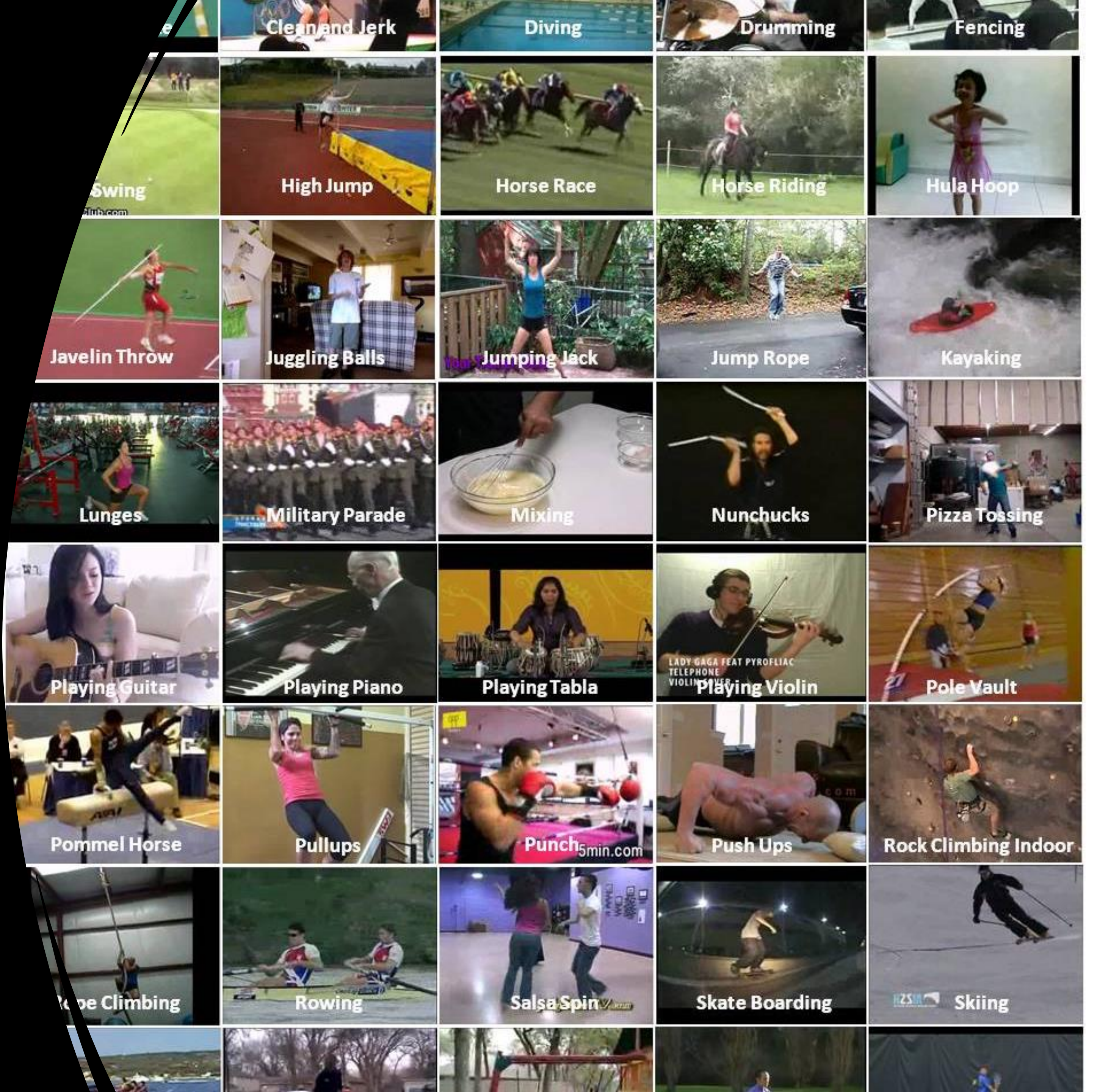


Related Work

Approach	Description	Limitations
Single-frame CNNs	Classifies each frame independently using CNN	Ignores temporal context – leads to misclassifications
Fusion Methods	Combine predictions across frames (Early/Late Fusion)	Limited temporal understanding
3D CNNs	Learn spatial-temporal features jointly across frames	High computational cost, memory-intensive
Pose + RNNs	Track human keypoints → feed into LSTM	Misses scene and object context

Dataset

50 diverse real-world human activity classes



Preprocessing

Frame Sampling

Sample 20 evenly spaced frames per video (`SEQUENCE_LENGTH = 20`)

Resizing

Resize each frame to 64×64 pixels for faster computation

Augmentation Techniques

- Random horizontal flip
- Random rotation

Tensor Transformation

- Convert to shape: `[T, C, H, W]`
- Normalize and batch using PyTorch DataLoaders

Proposed Solution

Long-term Recurrent Convolutional Network (LRCN)

Combines **CNN** (spatial features) + **LSTM** (temporal sequence)

CNN – Spatial Feature Extraction

- 2 convolutional layers (32 & 64 filters)
- Output: feature maps from each frame

LSTM – Temporal Modeling

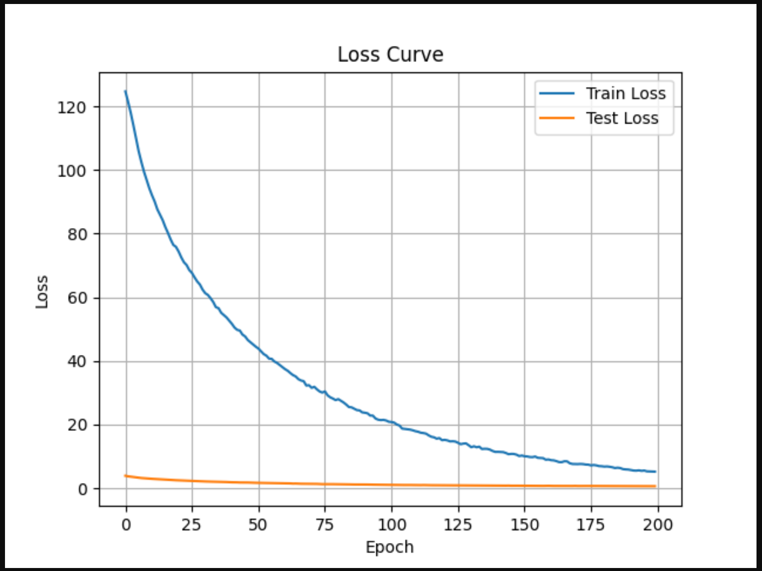
- 128 hidden units

Fully Connected Layer

- Maps final LSTM output to 50 action classes

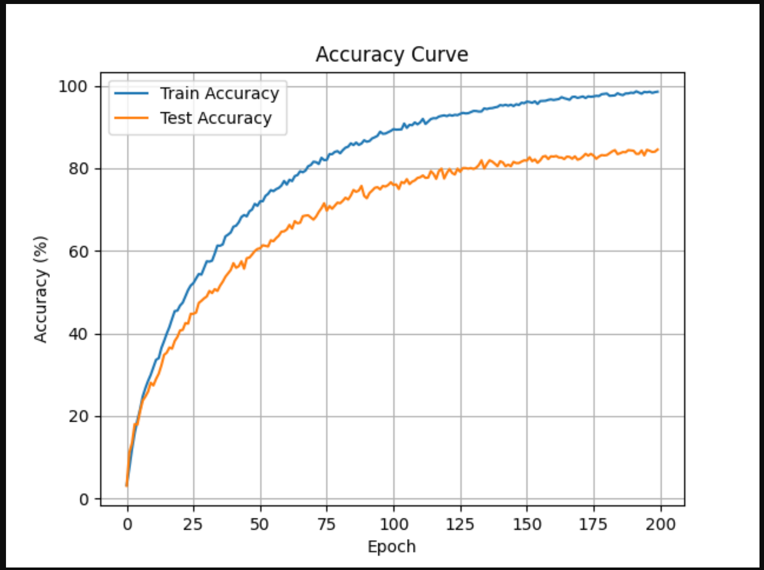
Training & Prediction

- **Loss Function**: CrossEntropyLoss
- **Optimizer**: Adam (learning rate = 0.0001)
- **Data Split**:
- 70% Training
- 10% Validation
- 20% Testing



Accuracy

- Final Train Accuracy: 98.85%
- Final Validation Accuracy: 84.43%
- Final Test Accuracy: 83.77%



Post-Training Analysis

- The Heatmap for Javelin Throw: https://youtu.be/O9gfeRcOvqk

Conclusion

- Successfully built a Human Activity Recognition (HAR) system using **LRCN** (CNN + LSTM)
- Achieved **83.77% accuracy** on test real-world video data (UCF50)
- Deployed as a **Flask web application** for real-time user interaction

Strengths of the System

- Combines **spatial** and **temporal** modelling effectively
- Lightweight and suitable for **real-time inference**
- Modular design: easy to extend or improve

DEMO

http://127.0.0.1:5000/